

No. of Functional Features Included in the Solution

Date	01 July 2026
Team ID	
Project Name	RISING WATERS
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Components	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Allows users to enter flood-Related problems	Frontend (HTML,CSS, Flask)	Done	10%
2	Data Preprocessing	Cleans and prepares the dataset before processing	Pandas, NumPy	Done	15%
3	Machine Learning Model	Trains the XGBoost model to predict the result	Scikit-learn, XGBoost	Done	20%
4	Flood Prediction	Predicts flood risk using the trained ML model	Prediction Engine	Done	20%
5	Prediction Result display	Displays the prediction result to the user	Flask Backend	Done	10%
6	Model Evaluation	Calculates model accuracy and evaluates model	Scikit-learn Metrics	Done	10%
7	Dataset Management	Loads and manages the flood dataset	Dataset Module	Done	10%
8	Error Handling	Handles invalid or incomplete values gracefully	Flask Validation	Done	5%

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Prediction Result Page
2	Backend / Logic	3	Data Preprocessing, Model Prediction, Error Handling
3	Database / Storage	1	Flood Dataset Storage
4	API / Integration	1	Flask Prediction API
5	Security / Authentication	1	Input Validation